

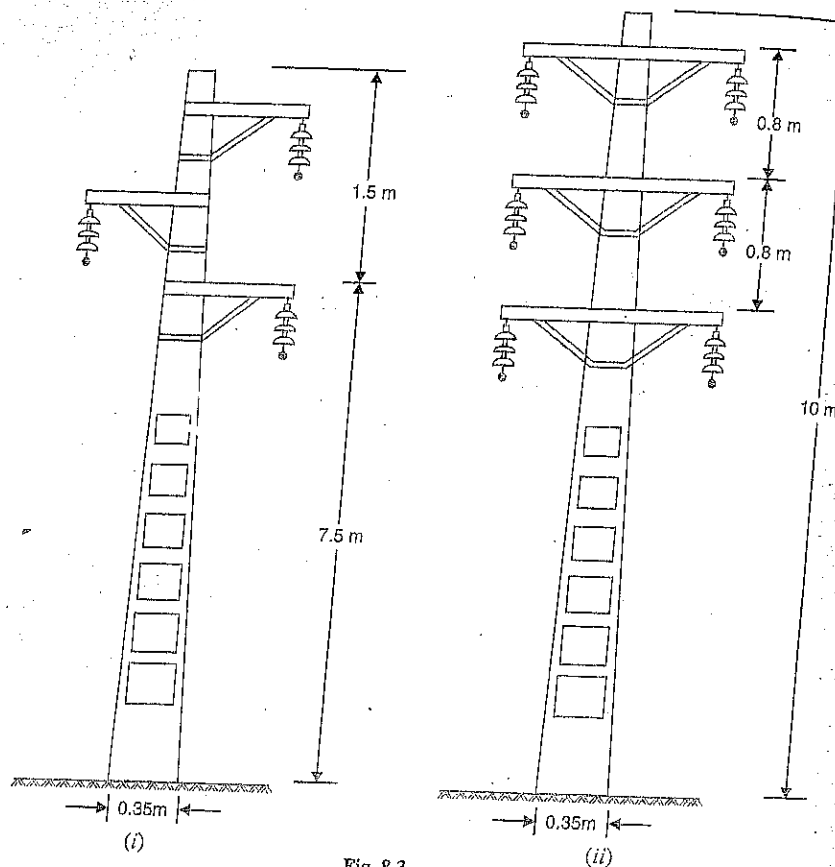


Fig. 8.3

life, can withstand most severe climatic conditions and permit the use of longer spans. The risk of interrupted service due to broken or punctured insulation is considerably reduced owing to longer spans. Tower footings are usually grounded by driving rods into the earth. This minimises the lightning troubles as each tower acts as a lightning conductor.

Fig. 8.4 (i) shows a single circuit tower. However, at a moderate additional cost, double circuit tower can be provided as shown in Fig. 8.4 (ii). The double circuit has the advantage that it ensures continuity of supply. If there is breakdown of one circuit, the continuity of supply can be maintained by the other circuit.

8.4 Insulators

The overhead line conductors should be supported on the poles or towers in such a way that currents from conductors do not flow to earth through supports i.e., line conductors must be properly insulated from supports. This is achieved by securing line conductors to supports with the help of insulators. The insulators provide necessary insulation between line conductors and supports and thus prevent any leakage current from conductors to earth. In general, the insulators should have the following desirable properties :

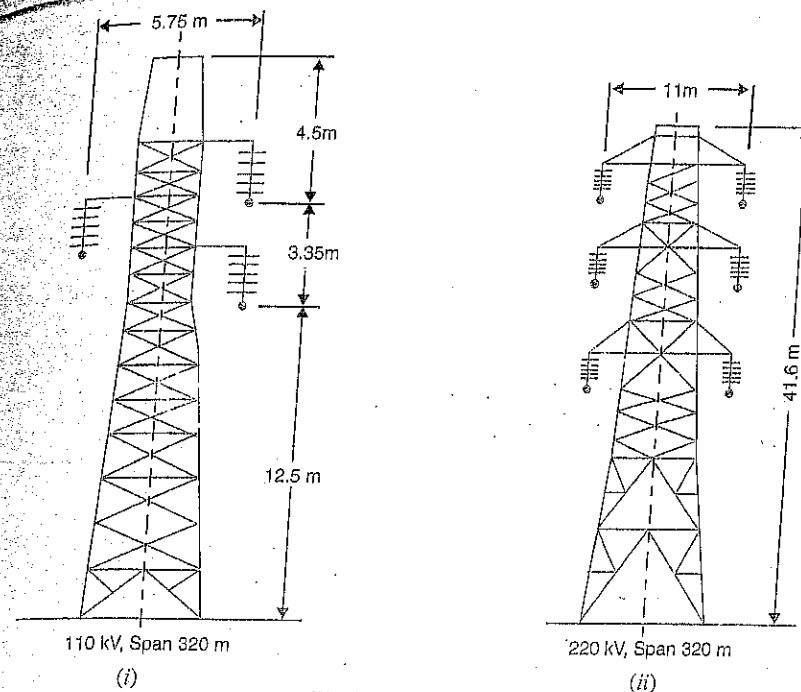


Fig. 8.4 Steel Towers

- High mechanical strength in order to withstand conductor load, wind load etc.
- High electrical resistance of insulator material in order to avoid leakage currents to earth.
- High relative permittivity of insulator material in order that dielectric strength is high.
- The insulator material should be non-porous, free from impurities and cracks otherwise the permittivity will be lowered.
- High ratio of puncture strength to flashover.

The most commonly used material for insulators of overhead line is *porcelain* but glass, steatite and special composition materials are also used to a limited extent. Porcelain is produced by firing at a high temperature a mixture of kaolin, feldspar and quartz. It is stronger mechanically than glass, gives less trouble from leakage and is less effected by changes of temperature.

Types of Insulators

The successful operation of an overhead line depends to a considerable extent upon the proper selection of insulators. There are several types of insulators but the most commonly used are pin type, suspension type, strain insulator and shackle insulator.

- Pin type insulators. The part section of a pin type insulator is shown in Fig. 8.5 (i). As the name suggests, the pin type insulator is secured to the cross-arm on the

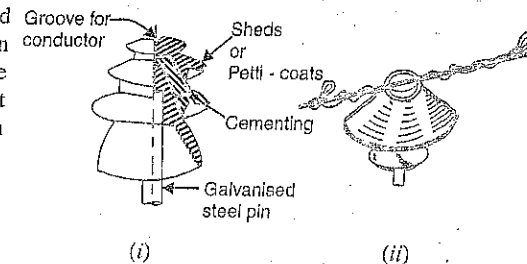


Fig. 8.5. Pin-type insulator

pole. There is a groove on the upper end of the insulator for housing the conductor. The conductor passes through this groove and is bound by the annealed wire of the same material as the conductor [See Fig. 8.5 (ii)].

Pin type insulators are used for transmission and distribution of electric power at voltages upto 33 kV. Beyond operating voltage of 33 kV, the pin type insulators become too bulky and hence uneconomical.

Causes of insulator failure. Insulators are required to withstand both mechanical and electrical stresses. The latter type is primarily due to line voltage and may cause the breakdown of the insulator. The electrical breakdown of the insulator can occur either by *flash-over* or *puncture*. In flash-over, an arc occurs between the line conductor and insulator pin (i.e., earth) and the discharge jumps across the air gaps, following shortest distance. Fig. 8.6 shows the arcing distance (i.e. $a + b + c$) for the insulator. In case of flash-over, the insulator will continue to act in its proper capacity unless extreme heat produced by the arc destroys the insulator.

In case of puncture, the discharge occurs from conductor to pin through the body of the insulator. When such breakdown is involved, the insulator is permanently destroyed due to excessive heat. In practice, sufficient thickness of porcelain is provided in the insulator to avoid puncture by the line voltage. The ratio of puncture strength to flash-over voltage is known as safety factor i.e.,

$$\text{Safety factor of insulator} = \frac{\text{Puncture strength}}{\text{Flash-over voltage}}$$

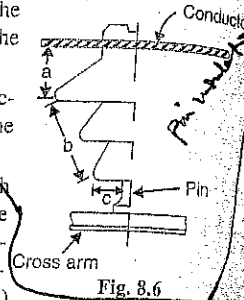


Fig. 8.6

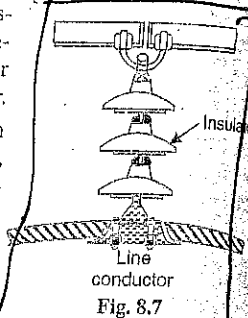
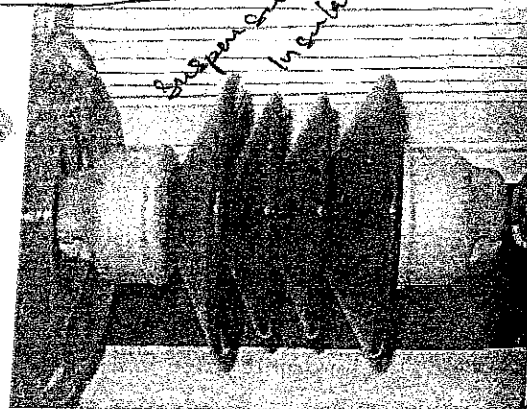


Fig. 8.7



Pin type Insulator



Suspension Insulator

It is desirable that the value of safety factor is high so that flash-over takes place before the insulator gets punctured. For pin type insulators, the value of safety factor is about 10.

Suspension type insulators. The cost of pin type insulator increases rapidly as the working voltage is increased. Therefore, this type of insulator is not economical beyond 33 kV. For high voltages (>33 kV), it is a usual practice to use suspension type insulators shown in Fig. 8.7. They

* The insulator is generally dry and its surfaces have proper insulating properties. Therefore, arc can only occur through air gap between conductor and insulator pin.

consist of a number of porcelain discs connected in series by metal links in the form of a string. The conductor is suspended at the bottom end of this string while the other end of the string is secured to the cross-arm of the tower. Each unit or disc is designed for low voltage, say 11 kV. The number of discs in series would obviously depend upon the working voltage. For instance, if the working voltage is 66 kV, then six discs in series will be provided on the string.

Advantages

- (i) Suspension type insulators are cheaper than pin type insulators for voltages beyond 33 kV.
- (ii) Each unit or disc of suspension type insulator is designed for low voltage, usually 11 kV. Depending upon the working voltage, the desired number of discs can be connected in series.
- (iii) If any one disc is damaged, the whole string does not become useless because the damaged disc can be replaced by the sound one.
- (iv) The suspension arrangement provides greater flexibility to the line. The connection at the cross arm is such that insulator string is free to swing in any direction and can take up the position where mechanical stresses are minimum.
- (v) In case of increased demand on the transmission line, it is found more satisfactory to supply the greater demand by raising the line voltage than to provide another set of conductors. The additional insulation required for the raised voltage can be easily obtained in the suspension arrangement by adding the desired number of discs.
- (vi) The suspension type insulators are generally used with steel towers. As the conductors run below the earthed cross-arm of the tower, therefore, this arrangement provides partial protection from lightning.

Strain insulators. When there is a dead end of the line or there is corner or sharp curve, the line is subjected to greater tension. In order to relieve the line of excessive tension, strain insulators are used. For low voltage lines (< 11 kV), shackle insulators are used as strain insulators. However, for high voltage transmission lines, strain insulator consists of an assembly of suspension insulators as shown in Fig. 8.8. The discs of strain insulators are used in the vertical plane. When the tension in lines is exceedingly high, as at long river spans, two or more strings are used in parallel.

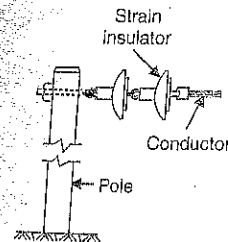


Fig. 8.8. Strain insulator.

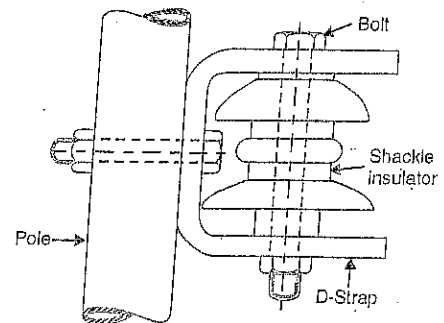


Fig. 8.9

Shackle insulators. In early days, the shackle insulators were used as strain insulators. But now a days, they are frequently used for low voltage distribution lines. Such insulators can be used either in a horizontal position or in a vertical position. They can be directly fixed to the pole with a bolt or to the cross arm. Fig. 8.9 shows a shackle insulator fixed to the pole. The conductor in the groove is fixed with a soft binding wire.

Tests on Insulators

Major tests carried out on insulators are:

1. type test 2. sample tests 3. routine tests

3.45.1

Type Tests

These tests are done to determine whether the particular design is suitable for the purpose, and they are as follows;

(a) Withstand tests: the insulator should be mounted so as to simulate practical conditions. $\frac{1}{50}$ MS wave of specified voltage (corrected for humidity, air density, etc) is applied. Flashover or puncture should not occur. [If puncture occurs, the insulator is permanently damaged]. The test is repeated five times for each polarity.

(b) Flashover tests: A $\frac{1}{50}$ MS wave is applied. The voltage is gradually increased to the 50% impulse flashover voltage. The test is done for both polarities. There should be no flashover or puncture of insulation during these tests.

(c) Dry one-minute test: the insulator, clean and dry, shall be mounted as specified and the prescribed voltage (corrected for ambient conditions) should be for one minute. (at power frequency) and maintained insulator. There should be no damage to the insulator.

(d) One-minute rain test: the insulator is sprayed throughout the test with artificial rain drawn from a source of supply at a temperature of 10°C of the ambient temperature of the neighbourhood of the insulator. The

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rain is sprayed at an angle of 45° on the insulator at the prescribed rate of 3mm/minute. The resistivity of the water should be $100\text{-ohm}\cdot\text{m} \pm 10\%$. The prescribed voltage is maintained for one minute and there should be no damage to the insulator.

(E) Wet flashover test: the ^{applied} voltage shall then be increased gradually until flashover occurs. This is repeated ten times. There shall be no damage to the insulator.

(F) Visible discharge test: this test starts after the room has been darkened and the specified test voltage applied, after five minutes, there should be no visible signs of corona, otherwise the insulator ~~cannot~~ is ~~not~~ suitable for the application.

3-15-2
3-4-2 Sample Tests: the various sample tests are carried out by the manufacturers before the insulator is dispatched to the market.

3-15-3
3-4-3 (a) Routine Tests: these are to be applied to all insulators and shall be commenced at a low voltage and shall be increased rapidly until flashover occurs every few seconds. The voltage shall be maintained at this value for a maximum of five minutes or if failure occurs, for five minutes after the last punctured piece has been removed. At the conclusion of the test the voltage before switching off.

(b) Mechanical Routine test: a mechanical load of 20% in excess of the maximum working load of the insulator is applied after suspending the insulator for one minute. There should be no mechanical failure of the insulator.

Voltage Distribution Across Suspension-Insulator String

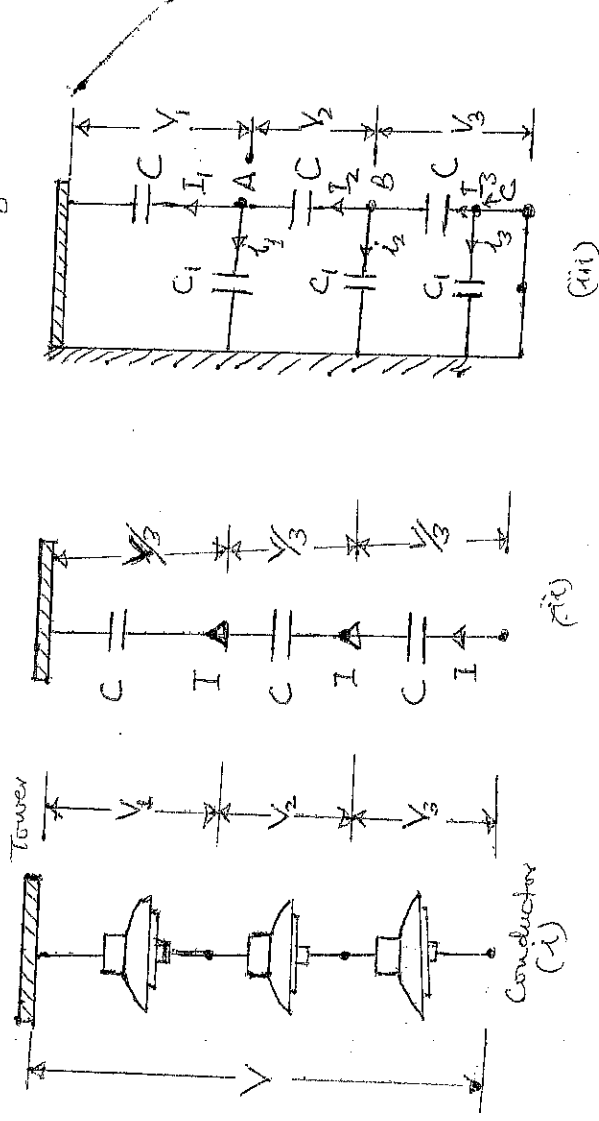


Fig. 3.5 (Voltage distribution across suspension insulator)

A string of suspension insulators consists of porcelain discs connected in series through ~~metallic~~ metallic links. Fig. 3.5(i) shows 3-disc string suspension insulators. The porcelain portion of each disc is in between two metal links. Therefore, each disc forms a capacitor C as shown in Fig. 3.5(ii).

This is known as "mutual capacitance" or "self capacitance". If there were mutual capacitance alone, then charging current would have been the same through all the discs and consequently voltage across each unit would have been same i.e., $V/3$ as shown in Fig. 3.5(iii). However, in actual practice, capacitance also exists between metal fitting of each disc and tower (or earth). This is known as "shunt capacitance" C_1 . Due to shunt capacitance, charging current is not the same through all the discs of the string (see Fig. 3.5(iii)). Therefore, voltage across each disc will be different. Obviously, the disc nearest to the line conductor will have the maximum voltage.

3.17 ~~3.16~~ Referring to Fig. 3.5(iii), V_3 will be much more than V_2 or V_1 .

Expression for Voltage Distribution Across Insulation String

From Fig 3.5(iii), the self capacitance of each disc is C while the shunt capacitance is C_1 and is some fraction K of self capacitance i.e., $C_1 = KC$. Starting from the cross-arm or tower, the voltage across each unit is V_1 , V_2 and V_3 respectively as shown.

Applying Kirchhoff's current Law to node A, we get

$$I_2 = I_1 + I_3$$

$$\text{or } V_2 WC = V_1 WC + V_1 WC_1$$

$$= V_1 WC + V_1 WK C$$

Dividing both sides by WC gives.

$$V_2 = V_1(1+K)$$

Applying Kirchhoff's current law to node B, we get

$$I_3 = I_2 + I_1$$

$$\text{or } V_3 WC = V_2 WC + (V_1 + V_2) WC_1$$

$$\text{or } V_3 WC = V_2 WC + (V_1 + V_2) WK C$$

$$\text{or } V_3 = V_2 + (V_1 + V_2) K$$

$$V_3 = V_2 + KV_1 + KV_2$$

$$V_3 = KV_1 + V_2(1+K)$$

From eqn ① $V_2 = V_1(1+K)$, then substituting for V_2 in expression for V_3 gives,

$$V_3 = KV_1 + V_1(1+K)(1+K)$$

(68) (82)

$$\text{or } V_3 = KV_1 + V_1(1+K)^2$$

$$\text{or } V_3 = KV_1 + V_1(1+2K+K^2)$$

$$V_3 = V_1[K+1+2K+K^2]$$

$$V_3 = V_1[1+3K+K^2]$$

----- (2)

Voltage between conductor and earth (i.e., tower) is

$$V = V_1 + V_2 + V_3$$

$$= V_1 + V_1(1+K) + V_1(1+3K+K^2)$$

$$= V_1(1+1+K+1+3K+K^2)$$

$$= V_1[3+4K+K^2]$$

$$\therefore V = V_1(1+K)(3+K) \quad \text{----- (3)}$$

From expressions (1), (2) and (3), we get,

$$V_1 = \frac{V_2}{1+K} = \frac{V_3}{1+3K+K^2} = \frac{V}{(1+K)(3+K)}$$

Hence, voltage across the top disc, V_1 is given by,

$$V_1 = \frac{V}{(1+K)(3+K)} \quad \text{----- (4)}$$

3.18
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String Efficiency

The voltage applied across the string of suspension insulators is not uniformly distributed across various units or discs. The disc nearest to the conductor has much higher potential than the other discs. This unequal potential distribution is undesirable and is usually expressed in terms of "string efficiency".

String efficiency is defined as the ratio of voltage across the string to the product of number of discs and the voltage

$$(3.18) (8.5)$$

across the disc nearest to the conductor is known as string efficiency i.e.,

$$\text{String Efficiency} = \frac{\text{Voltage Across the String}}{n \times \text{Voltage Across Disc Nearest to Conductor}}$$

$$= \frac{V}{n \times \frac{V}{3}} \times 100\%$$

where n = number of discs in the string.

Note: String efficiency is an important consideration since it decides the potential distribution along the string. The greater the string efficiency, the more uniform is the voltage distribution. Thus 100% string efficiency is an ideal case for which the voltage across each disc will be exactly the same. Although it is impossible to achieve 100% string efficiency, yet efforts should be made to improve it as close to this value as possible.

Example:

In a 33KV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self-capacitance of the three units, find (a) the distribution of voltage over the three units (b) the string efficiency.

Solution

$$K = \frac{\text{Shunt Capacitance}}{\text{Self - Capacitance}} = 0.11$$

Voltage across string, $V = \frac{33\text{KV}}{\sqrt{3}} = 19.05\text{KV}$

~~(7A)~~ (50)

Referring to Fig. 3.5(ii) above,

At junction A,

$$I_2 = I_1 + i_1$$

$$\text{or } V_2 \omega C = V_1 \omega C + V_1 k \omega C$$

$$\text{or } V_2 = V_1(1+k) = V_1(1+0.11)$$

$$V_2 = 1.11V_1$$

At junction B,

$$I_3 = I_2 + i_2$$

$$V_3 \omega C = V_2 \omega C + (V_1 + V_2) k \omega C$$

$$V_3 = V_2 + (V_1 + V_2) k$$

$$= 1.11V_1 + (V_1 + 1.11V_1) 0.11$$

$$V_3 = 1.342V_1$$

(a) Voltage across the three units; ②

$$V = V_1 + V_2 + V_3 = V_1 + 1.11V_1 + 1.342V_1 = 3.452V_1$$

$$\text{or } 19.05 \text{ kV} = 3.452V_1$$

$\therefore$ Voltage across top unit, $V_1 =$

$$\text{Voltage across top unit, } V_1 = \frac{19.05 \text{ kV}}{3.452} = \underline{\underline{5.52 \text{ kV}}}$$

$$\text{Voltage across middle unit, } V_2 = 1.11V_1$$

$$= 1.11 \times 5.52 \text{ kV} = \underline{\underline{6.13 \text{ kV}}}$$

$$\text{Voltage across bottom unit, } V_3 = 1.342V_1$$

$$= 1.342 \times 5.52 \text{ kV}$$

$$= \underline{\underline{7.4 \text{ kV}}}$$

$$\begin{aligned}
 \text{(b) String Efficiency} &= \frac{\text{Voltage Across String}}{\text{Number of Insulators} \times V_3} \times 100\% \\
 &= \frac{19.05 \text{ KV}}{3 \times 7.4 \text{ KV}} \times 100\% \\
 &= \underline{\underline{85.8\%}}
 \end{aligned}$$

Insulation Coordination

IEEE defines insulation coordination as the selection of insulation strength consistent with expected overvoltages to obtain an acceptable risk of failure. The procedure for insulation coordination consists of;

- determination of the voltage stresses, and
- selecting of an insulation strength to achieve the desired probability of failure.

The voltage stresses can be reduced by the application of surge-protective devices, switching device insertion resistors and controlled closing, shield wires, and improved grounding.

3.20

Impulse Generator

Fig. 3-20 shows the Marx's circuit used as an impulse generator. An impulse generator produces a uni-directional short time called the front time (t_f) and dies away more or less to zero in a comparatively greater time. The waveform falls from maximum value to half the peak value in a very long time called the tail time (t_t). The

square wave generator powers the circuit while the adjustments of resistors R_s , R_d and capacitors C_1 and C_2 are used to set the front time (t_f) and tail time (t_t) respectively.

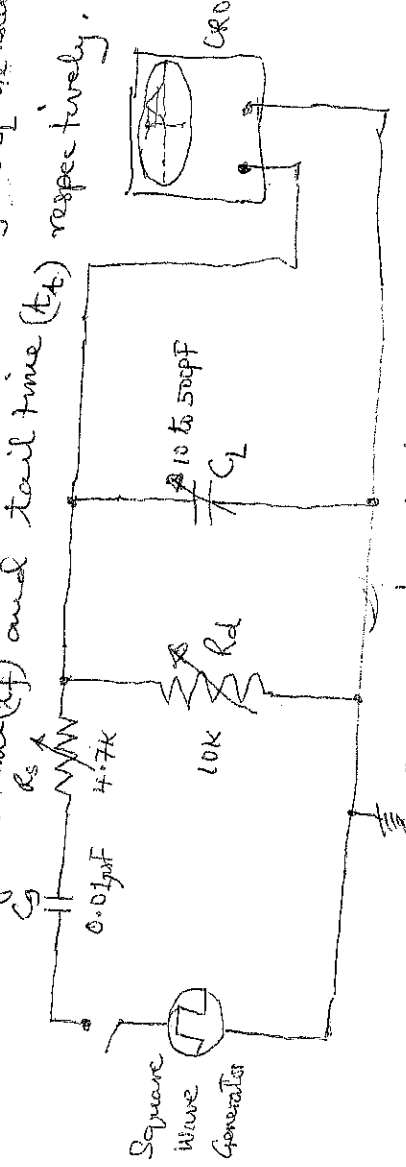


Fig. 3.20 Impulse Generator

(87)

3.2.1 ~~top~~ In order that equipment designed to be used on high voltage lines and others, be able to withstand surges caused in them during operation, it is necessary to test these equipment with voltages of the form likely to be met in service.

The apparatus which produces the required voltages is the impulse generator.

3.2.1

Impulse Generator Waveform

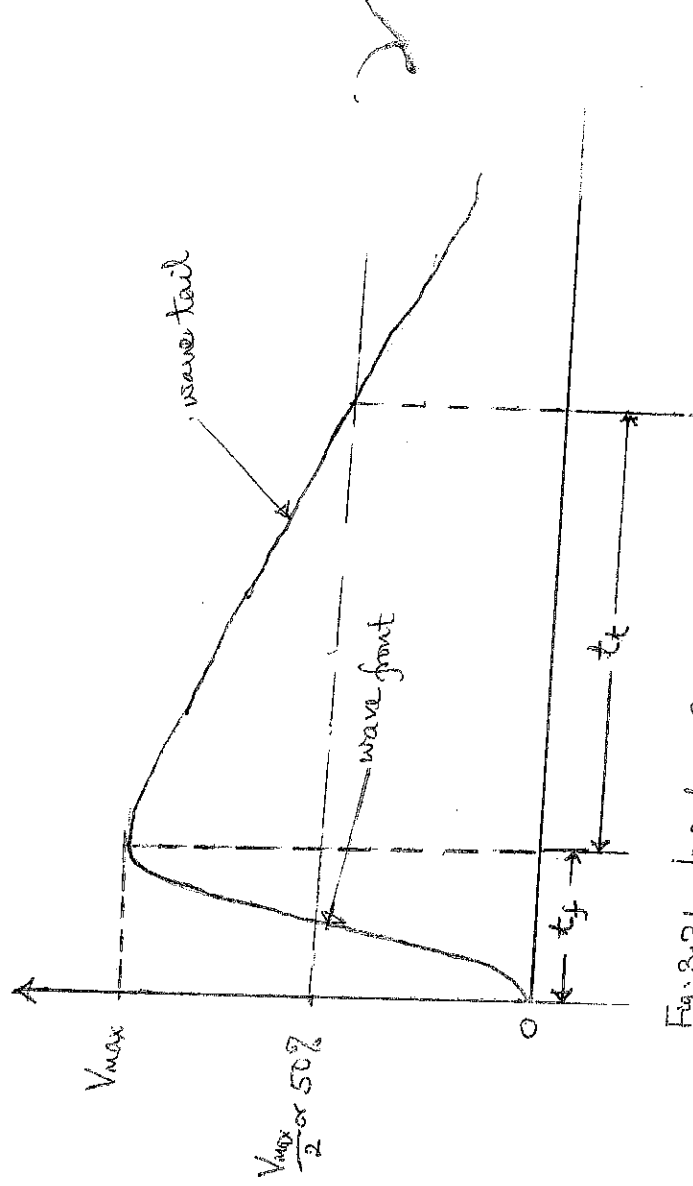
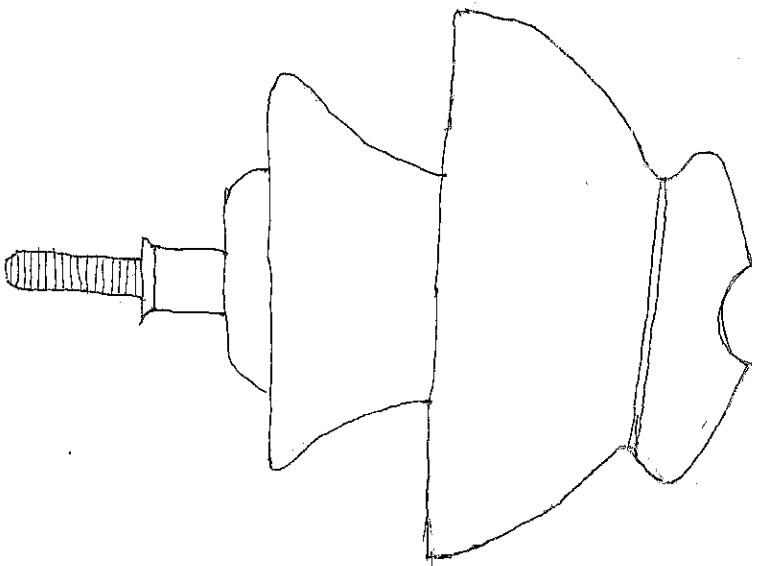


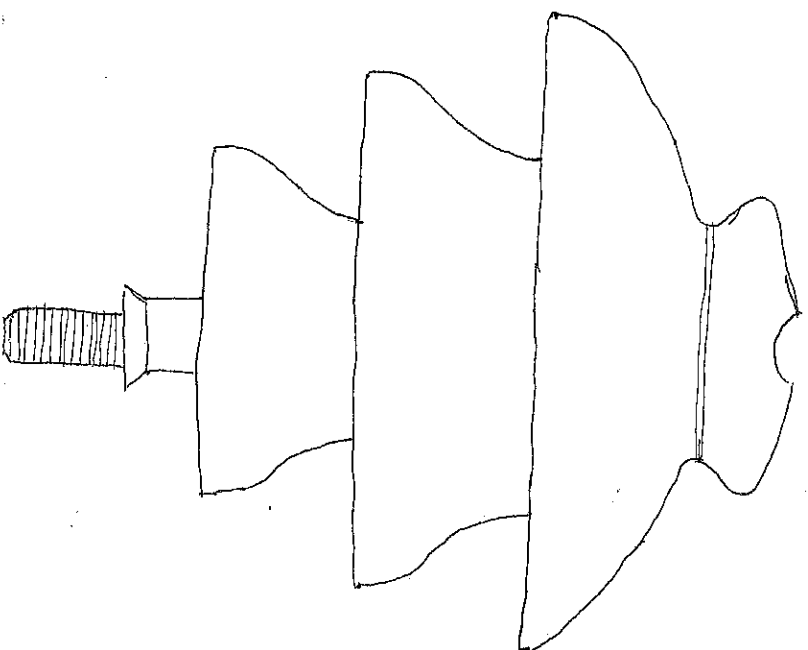
Fig. 3.2.1 Impulse Generator Waveform

The waveform obtained from an impulse generator is shown in Fig. 3.2.1. The output waveform is known as an impulse wave which is specified front time (t_f), the time it takes the wave to reach its peak value and the tail time (t_t), the time it takes the wave to fall from peak value to 50% of peak value. Typically, the front time is about 1 to 2 μ s while the tail time is about 50 μ s. The impulse waveform represents the shape of various surge voltages experienced in power systems.

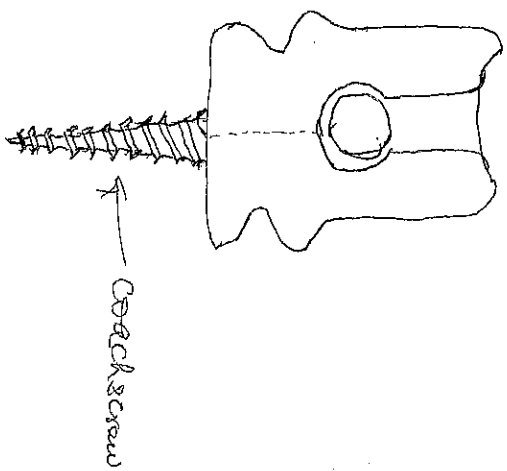
① P_n Mounted Type (11kV)



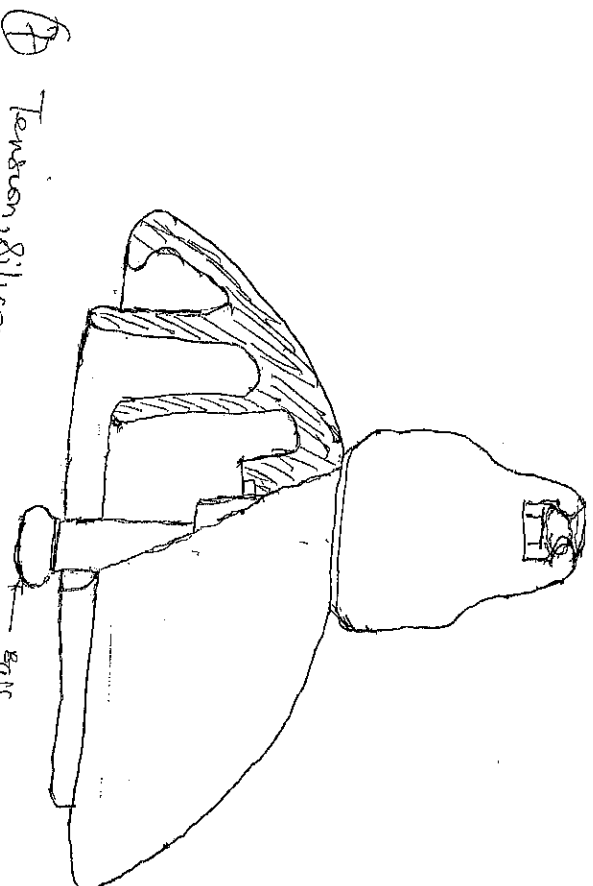
② P_n Mounted Type (38kV)



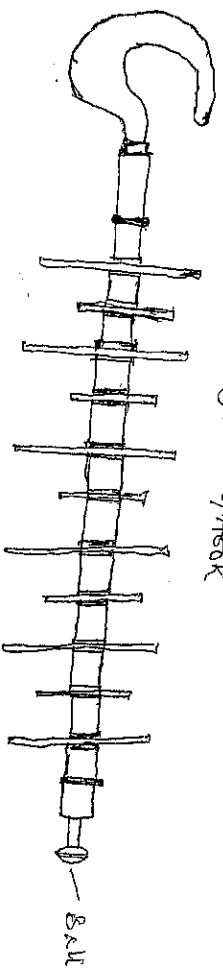
5) Coach Screw Service Type



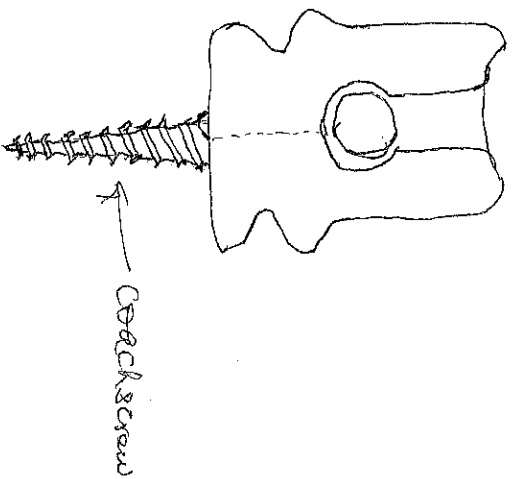
3) String Insulator Unit (Disc)



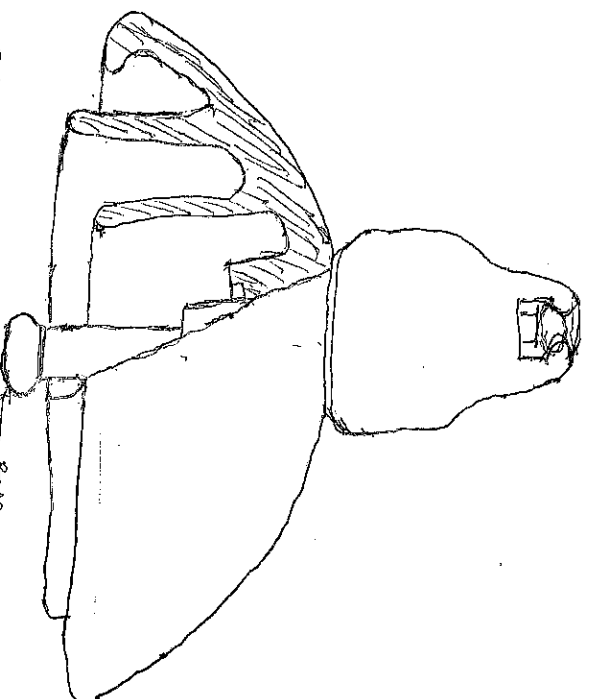
4) Tension, Silicone, Heavy duty, Ball/Hook



⑤ Cock Screw
Service Type



⑥ String Insulator Unit (Disc)



⑦ Tension, Sili cone, Heavy duty, Ball/Hook

